Evaluation Report

WordNet Similarity Measures for Legal Speech Analysis - Problem 15

Twisha | B.Tech IT (AI & ML Honors), SCET | Sem 6, 2026-27

1. Setup

The evaluation set contains **28 term pairs** (23 related legal pairs and 5 unrelated control pairs). Each pair uses one verified WordNet noun synset per term. Each pair has a relatedness rating from 0 (unrelated) to 4 (closely related).

Important caveat: the gold ratings were assigned by the project author, not by independent annotators, and the transcript is a short synthetic oral-argument excerpt written for this project (10 sentences, 40 legal-term occurrences, 30 distinct terms). Results show how the measures behave on this set; they are not a benchmark. Replacing gold_pairs.csv with ratings from several annotators would be the first improvement.

2. Agreement with gold ratings

metric	Spearman (Genesis)	95% CI (Spearman)	Pearson	Spearman (Brown)
Path	0.562	[0.18, 0.84]	0.498	0.562
Wu-Palmer	0.540	[0.19, 0.80]	0.505	0.540
Leacock-Chodorow	0.562	[0.18, 0.84]	0.530	0.562
Resnik	0.560	[0.24, 0.77]	0.525	0.575
Lin	0.614	[0.30, 0.80]	0.546	0.618

All five measures reach Spearman correlations between 0.54 and 0.61. Lin is highest (0.61 with Genesis IC, 0.62 with Brown IC), but with only 28 pairs the 95% bootstrap intervals overlap heavily, so the differences between metrics are **not statistically meaningful**. The reliable conclusion is that all measures capture a moderate monotone relationship with the gold ratings, not that one beats the others.

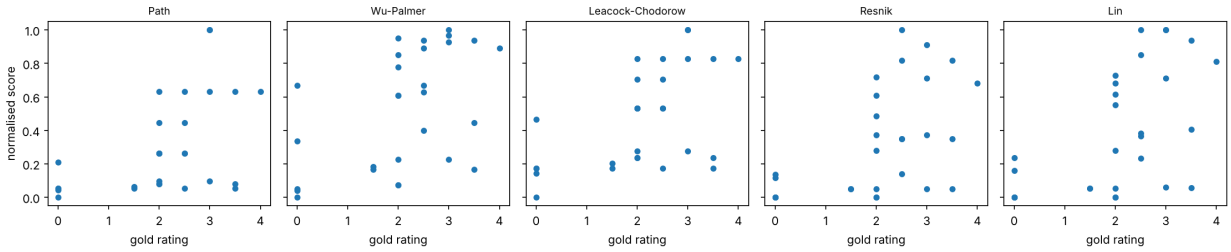


Figure 1. Normalised metric score against gold rating for every pair.

3. Related versus unrelated pairs

metric	related	control	gap
Path	0.349	0.072	0.277
Wu-Palmer	0.592	0.219	0.373
Leacock-Chodorow	0.530	0.191	0.339
Resnik	0.401	0.050	0.350
Lin	0.471	0.079	0.392

Every measure scores related pairs well above unrelated controls (gap 0.28 to 0.39 on the 0-1 normalised scale). Lin has the largest gap (0.39) and Path the smallest (0.28). Path scores are low overall even for good pairs because shortest-path length in WordNet is large for specific legal concepts.

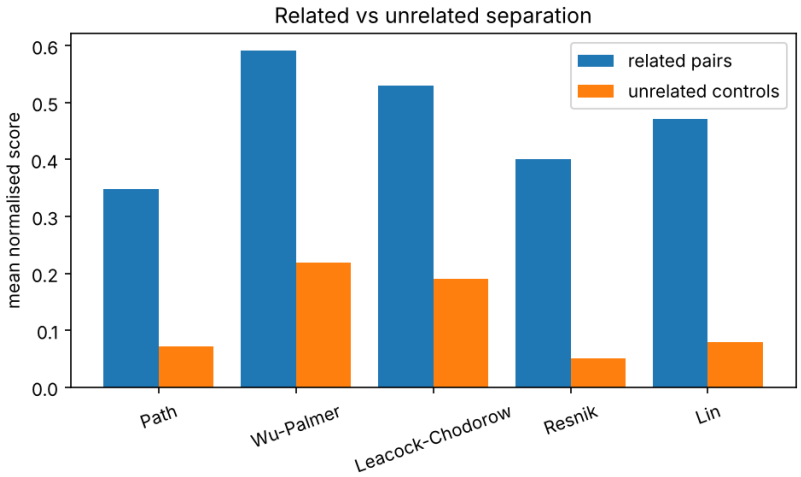


Figure 2. Mean normalised score on related pairs versus unrelated controls.

4. How the metrics relate to each other

	Path	Wu-Palmer	Leacock-Chodorow	Resnik	Lin
Path	1.000	0.938	1.000	0.832	0.850
Wu-Palmer	0.938	1.000	0.938	0.913	0.938
Leacock-Chodorow	1.000	0.938	1.000	0.832	0.850
Resnik	0.832	0.913	0.832	1.000	0.970
Lin	0.850	0.938	0.850	0.970	1.000

Path and Leacock-Chodorow have Spearman correlation 1.00: they rank every pair identically, as expected because LCH is a monotone transform of path length. Resnik and Lin are also highly correlated (0.97). The weakest relationship is between the path family and the IC family (Path-Resnik 0.83, Path-Lin 0.85), which is where the two approaches genuinely disagree.

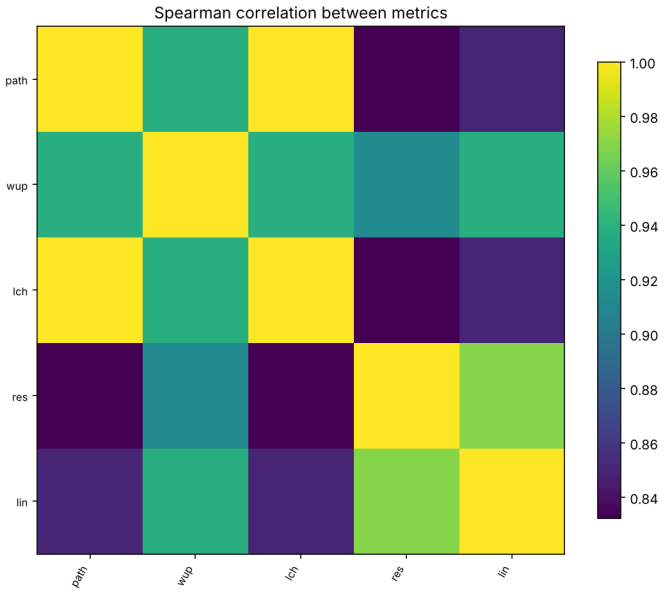


Figure 3. Spearman correlation between metrics.

5. Where taxonomic and information-theoretic measures disagree

term1	term2	WuP	Resnik	WuP rank	Resnik rank
lawyer	prosecutor	0.952	4.119	1	10
judge	prosecutor	0.909	4.102	3	11
oath	testimony	0.615	5.373	14.5	9
judge	dog	0.667	1.438	11.5	17
damages	fine	0.762	7.984	10	5

The table lists the pairs with the largest rank difference between Wu-Palmer and Resnik. Wu-Palmer ranks lawyer-prosecutor and judge-prosecutor near the top because the three are close in the person branch of the hierarchy. Resnik ranks them lower because their shared ancestor is a relatively general synset with modest information content. Conversely, Resnik rewards pairs such as damages-fine, whose shared ancestor is a specific concept about money, more than the path-based measures do. This is the central contrast of the problem: taxonomic measures reward *proximity in the tree*, information-theoretic measures reward *specificity of what is shared*.

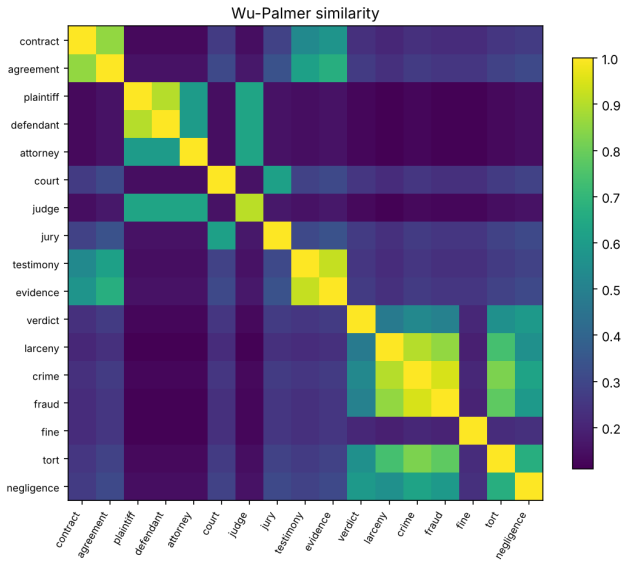


Figure 4. Wu-Palmer similarity matrix over 17 legal terms.

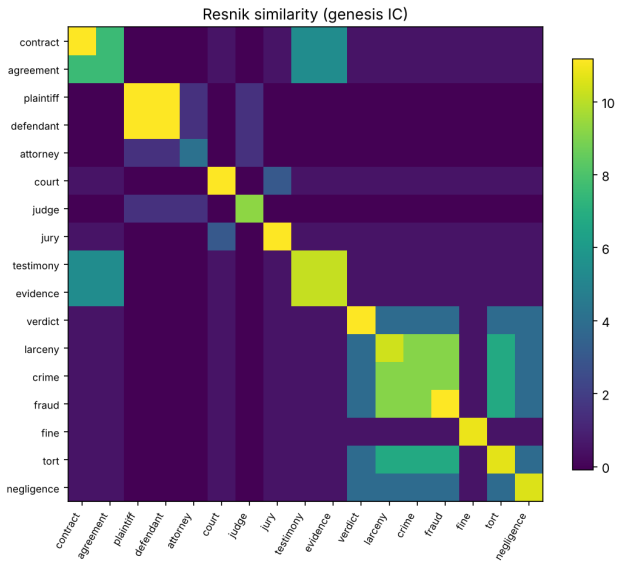


Figure 5. Resnik similarity matrix (Genesis IC).

6. Transcript-to-statute matching

Each distinct legal term in the transcript was matched to the closest of 10 statute terms with a normalised-score threshold of 0.6. Terms receiving a confident match: Path 10, Wu-Palmer 19, LCH 12, Resnik 15, Lin 15 of 30. Terms identical to a statute term score the maximum under every measure. Roles such as plaintiff, defendant, judge and attorney receive no confident match under any measure because the statute list contains no person terms; the threshold correctly refuses to force a match. Terms that do match well are mostly those whose statute counterpart is a near neighbour (e.g. damages to Fine, trial and appeal to Verdict).

7. Failure analysis

• **Similarity is not relatedness.** judge.n.01 and dog.n.01 receive a Wu-Palmer score of 0.67, higher than several legitimate legal pairs, because both are organisms and meet at organism.n.01, which is fairly deep (depth 5) relative to their own depths. Meanwhile court-judge, a strongly related pair, scores low on every measure because institution and person meet only at the root, entity.n.01. Wu-Palmer alone would be unreliable for filtering unrelated terms.

• **Path scores compress.** Even contract-agreement, a near-synonym pair, scores only 0.33 on Path (but 0.86 on Wu-Palmer), so absolute Path values are hard to threshold.

• **Negative information content.** Add-1 smoothing on the small Genesis corpus gives the root synset a slightly negative IC (about -0.083), so Resnik and Lin can be marginally negative for pairs that only share the root (e.g. plaintiff and contract). Brown IC returns exactly 0 there. Genesis is also biblical text, so its IC is a poor proxy for legal usage.

• **Hierarchy coverage.** Terms like 'statute' resolve to legislative_act.n.01, which sits far from law.n.01 in the hierarchy, so the pair law-statute scores low although a lawyer would treat them as close.

• **Sample size.** With 28 pairs and author-assigned ratings, correlations are imprecise (see the confidence intervals in Section 2).

8. Conclusions

• All five measures separate related legal pairs from unrelated controls and correlate moderately with the gold ratings; none clearly dominates on this evidence.

• Path and Leacock-Chodorow are rank-equivalent; Resnik and Lin behave very similarly. The meaningful comparison is between the path family and the information-content family.

• Taxonomic measures suit judging how close two concepts are in the hierarchy; information-content measures suit judging how specific the shared concept is. For legal text a combination, plus a legal-domain IC corpus and expert-rated pairs, would be a stronger design.

• Future work: build IC from a legal corpus, add multi-word terms and sense disambiguation, and evaluate against independent annotators.